

Data Base Lab
Lab Task 12



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Section 4A

Task 01

```
test> load("lab-14-devs.js")
true
lab_14_task> db.developers.find(
|   {
|     country: { $ne: "Pakistan" },
|     experience: { $gte: 3, $lte: 10 }
|   },
|   { name: 1, country: 1, experience: 1, _id: 0 }
| )
[
  { name: 'Sara', country: 'USA', experience: 7 },
  { name: 'John', country: 'UK', experience: 5 },
  { name: 'David', country: 'Canada', experience: 10 }
]
lab_14_task> |
```

Task 02

```
lab_14_task> db.developers.find({
|   $or: [
|     { skills: "JavaScript" },
|     {
|       experience: { $gt: 7 },
|       current_project: { $ne: null }
|     }
|   ]
| })
[
  {
    _id: ObjectId('69f8bf952f581d7929abc114'),
    name: 'Ali',
    country: 'Pakistan',
    experience: 3,
    skills: [ 'JavaScript', 'MongoDB', 'Node.js' ],
    current_project: { name: 'Ecommerce API', budget: 5000, status: 'active' }
  },
  {
    _id: ObjectId('69f8bf952f581d7929abc118'),
    name: 'David',
    country: 'Canada',
    experience: 10,
    skills: [ 'Go', 'Kubernetes', 'Docker' ],
    current_project: { name: 'Cloud Platform', budget: 30000, status: 'active' }
  }
]
lab_14_task> |
```

Task 03

```
lab_14_task> db.projects.find({
|   $and: [
|     { technologies: { $in: ["Node.js", "Docker"] } },
|     { budget: { $gt: 5000 } }
|   ]
| },
| { title: 1, domain: 1, budget: 1, _id: 0 }
| )
[ { title: 'Cloud Platform', domain: 'Cloud', budget: 30000 } ]
lab_14_task> |
```

Task 04

```
lab_14_task> db.projects.find({
|   $and: [
|     { team_size: { $gte: 2, $lte: 5 } },
|     { $expr: { $eq: [{ $size: "$technologies" }, 2] } }
|   ]
| })
[
|   {
|     _id: ObjectId('69f8bf952f581d7929abc119'),
|     title: 'Ecommerce API',
|     domain: 'Ecommerce',
|     technologies: [ 'Node.js', 'MongoDB' ],
|     budget: 5000,
|     team_size: 3
|   },
|   {
|     _id: ObjectId('69f8bf952f581d7929abc11a'),
|     title: 'Healthcare System',
|     domain: 'Healthcare',
|     technologies: [ 'Python', 'Django' ],
|     budget: 20000,
|     team_size: 5
|   },
|   {
|     _id: ObjectId('69f8bf952f581d7929abc11b'),
|     title: 'Banking App',
|     domain: 'Finance',
|     technologies: [ 'Java', 'Spring' ],
|     budget: 15000,
|     team_size: 4
|   }
| ]
lab_14_task> |
```

Task 05

```
lab_14_task> db.developers.find(
|   {
|     skills: { $ne: "MongoDB" },
|     experience: { $lte: 5 }
|   },
|   { name: 1, skills: 1, _id: 0 }
| ).limit(2)
|
| [
|   { name: 'John', skills: [ 'Java', 'Spring', 'MySQL' ] },
|   { name: 'Ayesha', skills: [ 'HTML', 'CSS' ] }
| ]
lab_14_task> |
```

Task 06

```
lab_14_task> db.developers.find({
|   $or: [
|     { current_project: null },
|     { $expr: { $eq: [{ $size: "$skills" }, 2] } }
|   ]
| },
| { name: 1, country: 1, skills: 1, current_project: 1, _id: 0 }
| )
|
| [
|   {
|     name: 'Ayesha',
|     country: 'Pakistan',
|     skills: [ 'HTML', 'CSS' ],
|     current_project: null
|   }
| ]
lab_14_task> |
```

Task 07

```
lab_14_task> db.developers.find(
|   { experience: { $gte: 3 } }
| ).sort({ experience: -1 }).limit(3)
[
  {
    _id: ObjectId('69f8bf952f581d7929abc118'),
    name: 'David',
    country: 'Canada',
    experience: 10,
    skills: [ 'Go', 'Kubernetes', 'Docker' ],
    current_project: { name: 'Cloud Platform', budget: 30000, status: 'active' }
  },
  {
    _id: ObjectId('69f8bf952f581d7929abc115'),
    name: 'Sara',
    country: 'USA',
    experience: 7,
    skills: [ 'Python', 'Django', 'PostgreSQL' ],
    current_project: { name: 'Healthcare System', budget: 20000, status: 'completed' }
  },
  {
    _id: ObjectId('69f8bf952f581d7929abc116'),
    name: 'John',
    country: 'UK',
    experience: 5,
    skills: [ 'Java', 'Spring', 'MySQL' ],
    current_project: { name: 'Banking App', budget: 15000, status: 'active' }
  }
]
lab_14_task> |
```

Task 8

```
lab_14_task> db.developers.find({
|   $and: [
|     { skills: { $in: [ "Python", "Go" ] } },
|     { skills: { $ne: "Java" } },
|     { experience: { $gt: 4 } }
|   ]
| },
| { name: 1, skills: 1, experience: 1, _id: 0 }
| )
[
  {
    name: 'Sara',
    experience: 7,
    skills: [ 'Python', 'Django', 'PostgreSQL' ]
  },
  {
    name: 'David',
    experience: 10,
    skills: [ 'Go', 'Kubernetes', 'Docker' ]
  }
]
lab_14_task> |
```

Task 9

```
lab_14_task> db.developers.find({
|   "current_project.budget": { $gt: 10000 },
|   "current_project.status": "active"
| },
| {
|   name: 1,
|   "current_project.name": 1,
|   "current_project.budget": 1,
|   _id: 0
| }
| )
| [
|   {
|     name: 'John',
|     current_project: { name: 'Banking App', budget: 15000 }
|   },
|   {
|     name: 'David',
|     current_project: { name: 'Cloud Platform', budget: 30000 }
|   }
| ]
lab_14_task> |
```

Task 10

```
lab_14_task> db.developers.find({
|   $and: [
|     { skills: { $all: ["JavaScript", "MongoDB"] } },
|     { "current_project.budget": { $gte: 4000, $lte: 10000 } }
|   ]
| },
| {
|   name: 1,
|   skills: 1,
|   "current_project.budget": 1,
|   _id: 0
| }
| )
| [
|   {
|     name: 'Ali',
|     skills: [ 'JavaScript', 'MongoDB', 'Node.js' ],
|     current_project: { budget: 5000 }
|   }
| ]
lab_14_task> |
```